



LiverNet: efficient and robust deep learning model for automatic diagnosis of sub-types of liver hepatocellular carcinoma cancer from H&E stained liver histopathology images

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Abstract

Purpose Liver cancer is one of the most common types of cancers in Asia with a high mortality rate. A common method for liver cancer diagnosis is the manual examination of histopathology images. Due to its laborious nature, we focus on alternate deep learning methods for automatic diagnosis, providing significant advantages over manual methods. In this paper, we propose a novel deep learning framework to perform multi-class cancer classification of liver hepatocellular carcinoma (HCC) tumor histopathology images which shows improvements in inference speed and classification quality over other competitive methods.

Method The BreastNet architecture proposed by Togacar et al. shows great promise in using convolutional block attention modules (CBAM) for effective cancer classification in H&E stained breast histopathology images. As part of our experiments with this framework, we have studied the addition of atrous spatial pyramid pooling (ASPP) blocks to effectively capture multi-scale features in H&E stained liver histopathology data. We classify liver histopathology data into four classes, namely the non-cancerous class, low sub-type liver HCC tumor, medium sub-type liver HCC tumor, and high sub-type liver HCC tumor. To prove the robustness and efficacy of our models, we have shown results for two liver histopathology datasets—a novel KMC dataset and the TCGA dataset.

Results Our proposed architecture outperforms state-of-the-art architectures for multi-class cancer classification of HCC histopathology images, not just in terms of quality of classification, but also in computational efficiency on the novel proposed KMC liver data and the publicly available TCGA-LIHC dataset. We have considered precision, recall, F1-score, intersection over union (IoU), accuracy, number of parameters, and FLOPs as metrics for comparison. The results of our meticulous experiments have shown improved classification performance along with added efficiency. LiverNet has been observed to outperform all other frameworks in all metrics under comparison with an approximate improvement of 2% in accuracy and F1-score on the KMC and TCGA-LIHC datasets.

Conclusion To the best of our knowledge, our work is among the first to provide concrete proof and demonstrate results for a successful deep learning architecture to handle multi-class HCC histopathology image classification among various sub-types of liver HCC tumor. Our method shows a high accuracy of 90.93% on the proposed KMC liver dataset requiring only 0.5739 million parameters and 1.1934 million floating point operations per second.

Keywords Convolutional neural network · Deep learning · HCC histopathology images · Medical image classification · Liver cancer grading

The python code for the implementation of this can be found here:
<https://github.com/shyamfec/LiverNet>.

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Introduction

Liver cancer was the fourth highest leading cause of cancer deaths and sixth most common cancer worldwide as of 2018 [1]. It is defined as the abnormal change in deoxyribonucleic acid (DNA) of liver cells which makes them divide and grow rapidly, ultimately turning into a tumor. In 2018, it is estimated to have caused 781,631 deaths worldwide [2]. It is mostly occurring in the context of chronic liver diseases leading to cirrhosis caused by hepatitis B, hepatitis C, diabetes, and certain other hereditary diseases. A primary reason for deaths related to liver cancer is that it is often diagnosed at an intermediate or final stage which greatly reduces the chances of survival of the patient.

Hepatocellular carcinoma (HCC) is the most common histological type of primary liver cancer that develops in liver cells. In 2019, HCC related fatality was the fourth most common cause of cancer deaths worldwide [3]. Histopathological analysis helps in identifying benign or malignant lesions and estimating the histological sub-type of liver HCC tumors. Furthermore, histological grading is used as a scale for estimating tumor malignancies. Low sub-type liver HCC tumors, in most cases, have good chances of recovery, whereas high sub-type tumors indicate lower survival rates. The main standard used for determining the sub-type of HCC is the Edmondson and Steiner's standard. It defines the following four tumor sub-types: L1, L2, L3, and L4. The lowest sub-type is given to L1 and the highest to L4, and L2 and L3 occupy the sub-types in between. In our experiments, we have grouped sub-type L3 and L4 into a single sub-type called: high sub-type. The purpose of this is because of the inherent similarities in L3 and L4 in cell size, cell shape, and patient fatality. L2 corresponds to medium sub-type, and L1 corresponds to low sub-type of HCC. Hence, in our analysis, we have the following four classes: non-cancerous or normal (Class 0), low sub-type liver HCC tumor—L1 (Class 1), medium sub-type liver HCC tumor—L2 (Class 2), and high sub-type liver HCC tumor—L3 and L4 (Class 3). Examples from our dataset are depicted in Fig. 1.

Manual detection of liver cancer requires pathologists to examine microscopic tissues of histopathology biopsy images. The conventional method of detection of Liver cancer is by staining liver tissue with hematoxylin and eosin

stains (H&E stain) which is then analyzed under a microscope by a pathologist. Hematoxylin stains the nuclei blue, and eosin stains the cytoplasm and extracellular matrix pink. A pathologist looks for different features in the cell such as the size of nuclei, stained nuclei color, distribution of the nuclei, and the shade of the cytoplasm due to stain. Cancerous tissues have distinct features such as an increased distribution and decentralization of nuclei and a darker stained cytoplasm, which become more intense in later stages of cancer. Taking such features into account, a pathologist comments about the cancer and stage it is in.

Liver cancer detection has always been a challenging task for pathologists and medical practitioners. The manual identification of liver cancer from histopathology images is time-consuming, laborious, and subjective as results may vary based on the expertise and bias of the pathologist. For example, it is a challenging task for pathologists to diagnose low sub-type liver HCC tumors correctly because of uneven tumor presence and similar structural nature within different tumor sub-types. To assist pathologists, computer-aided detection (CAD) systems have been developed using image processing algorithms and deep learning methods have been proposed as an alternative and efficient approach to alleviate the above-mentioned problems in manual detection. They can be seen as the diagnosis methods of the digital age that can carry out the diagnosis procedure with minimal error and fatigue. The main objective of this research study is to design and develop an efficient model for the fully automated detection and classification of HCC into four sub-types from H&E stained HCC histopathology images and improve upon the standard detection practice of each stage with more accurate and efficient classifications. The various stages involved in the proposed system include preprocessing, feature extraction, and classification. The contributions of the manuscript are as follows:

- (1) A novel architecture, called LiverNet, is to perform improved automatic diagnosis of sub-types of HCC from H&E stained liver histopathology images.
- (2) A dataset of stained HCC liver tissue obtained from Kasturba Medical College(KMC), Mangalore, Manipal Academy of Higher Education(MAHE), Manipal, Karnataka, India, has been collated and made open for public use. In this paper, we refer to this dataset as the KMC dataset.
- (3) Experimental results of the proposed architecture have been shown to outperform previous state-of-the-art deep learning architectures on the new dataset (KMC dataset) and on the existing TCGA-Liver dataset in terms of quality of classification and in computational efficiency.

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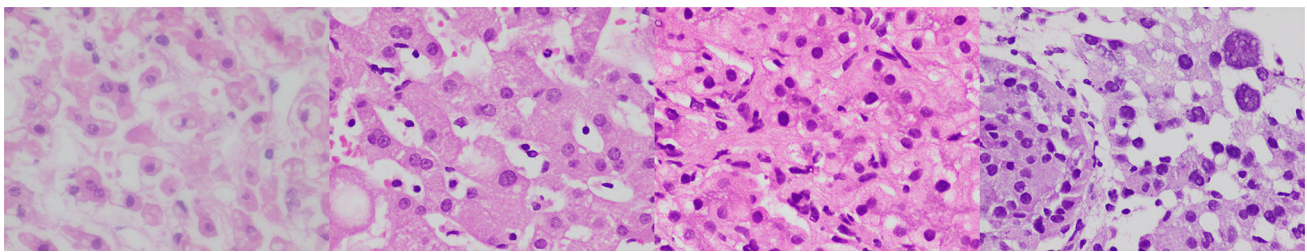


Fig. 1 Test data images for each class (normal, low sub-type liver HCC tumor, medium sub-type liver HCC tumor, high sub-type liver HCC tumor) from left to right

The organization of the manuscript is as follows: In “Related work” section, we elaborate on the related work in this domain. In “Proposed architecture” section, we described the proposed architecture. Implementation and training are presented in “Implementation and training” section. We present our results and comparisons in “Results” section, and in “Conclusion” section, we conclude our proposed work. The methodology that has been used to train the network can be observed in Fig. 2.

Related work

The previous decade saw remarkable advances in diagnostic imaging techniques. The development of sophisticated robotic microscopes allowed capturing and storing high-resolution H&E stained tissue images. With the advent of pattern recognition and machine learning, the process of cancer detection became automatic and computerized. However, this automation revolution brought with it a set of challenges. Firstly, the size and density of histopathology data are huge compared to radio-graphic data and hence require strong computational resources for processing [4,5]. For instance, a biopsy procedure will require the processing and storage of billions of pixel data when compared to millions in radio-graphic data analysis [4,5].

In recent years, convolutional neural networks (CNN) have become the state-of-the-art algorithm for many computer vision tasks. The AlexNet architecture [6] proposed by Krizhevsky et al. revolutionized the use of CNNs for the image classification task. AlexNet made use of *ReLU* activation functions which was faster and more efficient than the earlier used *tanh* or *sigmoid* activation functions. To improve the performance of CNNs, Simonyan et al. proposed the use of small kernels of size 3x3 for convolutional layers [7]. These kernels helped in extracting finer features from the image. They also demonstrated the use batch normalization [8] through its benefits in training deeper layers without covariate shift.

The ResNet architecture [9] proposed by He et al. enabled deep networks to be trained effectively, avoiding the issue of

vanishing gradients through shortcut connections between the input and output of each convolutional block. Furthermore, they replaced dense layers with global average pooling in order to minimize over-fitting. Huang et al. [10] solved the issue of training very deep networks by modifying the shortcut connection and connecting every layer with each other [9]. This helped in dropping connections to layers that do not contribute much, and reducing the total number of parameters required. Google introduced its novel inception module in the architecture called GoogLeNet [11]. The inception module carried out convolution with a variety of kernel sizes to extract information at different scales and hence was able to generalize feature extraction for all varieties of images. Shortcut connections were added in the inception module to create the Inception-ResNet V2 architecture [12].

Although the performance in the image recognition task was improved using these state-of-art-the architectures, the parameters and other computational complexities were still very large for simple platforms. The MobileNet architecture [13] proposed by Andrew et al. optimized the conventional CNN architecture by using depthwise separable convolutions instead of standard vanilla convolutions. Depthwise separable convolution required 8 to 9 times fewer parameters for the same convolution operation, reducing the complexity of the architecture to a great extent. MobileNetV2 architecture [14] further improved the performance of MobileNet architecture by adding a shortcut connection between the input and output of depthwise separable convolutions.

Ferreira et al. [15] demonstrated that a pre-trained InceptionResNet-V2 could be used for the classification of breast cancer histopathology images through transfer learning. Nan et al. [16] presented a novel deep CNN architecture for improved breast cancer screening exam classification. The authors in [16] proposed a novel two-stage architecture with a custom ResNet-based design. The WSI-Net [17] proposed by Ni et al. processes entire whole slide images (WSI) in a forward pass by internally breaking up the slide into patches and fusion after segmentation. The architecture proposed by Chen et al. [18] makes use of the inception-V3 model in fusion with easy deep learning (DL) model for liver cancer prediction. The method of classifi-

cation of liver cancer as demonstrated in [19] carried out extensive preprocessing and WSI feature extraction before passing it to their model. Sun et al. [20] proposed a transfer learning-based classification algorithm into global classes. The authors demonstrated a feature extraction process using CNNs which could successfully classify the image data into normal and abnormal classes. Our proposed model, however, classifies the images into four sub-types of liver HCC tumor. Lu et al. [21] successfully extracted features from HCC histopathological images using VGG 16, Inception V3 and ResNet-50 and showed that there is a clear separation between the non-cancerous and cancerous features.

Chen et al. used the Inception-V3 architecture for automatic classification and mutation prediction of HCC histopathological images. For a similar application, Liao et al. [22] proposed a deep learning based using a patch-extraction process. Alom et al. [23] introduced Inception Recurrent Residual Convolutional Neural Network (IRRCNN) for classification of histopathology images. It made use of the recurrent CNN block concatenated with the inception block. Togcar et al. proposed BreastNet [24] architecture for breast cancer classification task. They introduced a channel attention unit and an spatial attention unit. They also made use of bilinear upsampling layer, which allowed the architecture to extract features from 3 different depths in the model. To the best of our knowledge, our proposed LiverNet model is among the first to establish a concrete methodology and results in the domain multi-grading HCC diagnosis.

Proposed architecture

At the heart of every CNN lies the convolution operation. Two-dimensional discrete linear convolution can be expressed as follows:

$$g[m, n] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l]h[m - k, n - l] \quad (1)$$

where f and h are two two-dimensional signals. Using this definition, we add the bias to Eq. (1) to obtain the computation formula per node in a given layer. In addition to this, another crucial operation in most CNN architectures today are max-pooling operations [6]. This operation can best be understood by considering a sliding window over the input feature map to the max-pool layer. This operation yields the maximum value of the pixels inside the window and is repeated across the whole image by sliding the window with a stride S . Max-pool layers help reduce the computational complexity of the network by reducing the number of parameters and help give an abstract representation of the input data.

A central problem faced in many computer vision tasks is the fact that objects of interest can appear in different scales depending on the scenario. Objects appearing in multiple scales have been a difficult task to handle for computer vision algorithms, and one of the popular approaches to address this problem was explored in [25,26]. A primary problem in pathological analysis of cancer tumors is that cancer causes abnormalities in cell size and shape. To solve this problem, we focus on varying scale features that correspond to these physical abnormalities. We intend to preserve this scale dependency to perform better classification between them. Chen et al. proposed the use of an atrous spatial pyramid pooling block after the encoder framework to extract feature-rich information at varying scales. This combines the idea of spatial pyramid pooling [26] to capture multi-scale features, and atrous convolutions which can efficiently increase the receptive field size without increasing the number of parameters involved. In [25], this was primarily applied for the task of image segmentation, but in our proposed architecture, we use it for the application of image classification.

We adopt the base architecture similar to [24], and we have used two convolution layers to extract features from the input image before the first max-pool operation. We have used CBAM blocks [27] and residual blocks deeper in the architecture to extract relevant information more effectively. We use intermediate features in the encoder pipeline after each max-pool operation to feed into ASPP blocks before upsampling. The hyper-column technique used in [24] was also adopted here to combine the pixel data of layers of varying depth. In the proposed framework, the hyper-column technique coupled with ASPP blocks ensures multi-scale feature extraction and information retrieval for further processing. In our proposed architecture, we have extended these ideas to the problem of multi-class cancer classification in liver tissue and a detailed depiction of the proposed model can be found in Fig. 3. The sub-modules of the proposed architecture have been described in detail in the following subsections.

CBAM block and residual block

The concept of convolutional block attention module (CBAM) was proposed by Woo et al. [27]. They demonstrated that the CBAM block can be efficiently inserted into any CNN architecture without too much overhead on computation and memory performance. They proposed the use of consecutive channel-wise and spatial attention modules to yield attention maps that can be multiplied against the input feature map. Woo et al. [27] demonstrated an improvement in classification performance when CBAM was included into state-of-the-art architectures. At the heart of our proposed model lie CBAM blocks that focus the attention of the network on objects of interest. The channel-wise attention block in CBAM pays attention to **what** the network must focus on

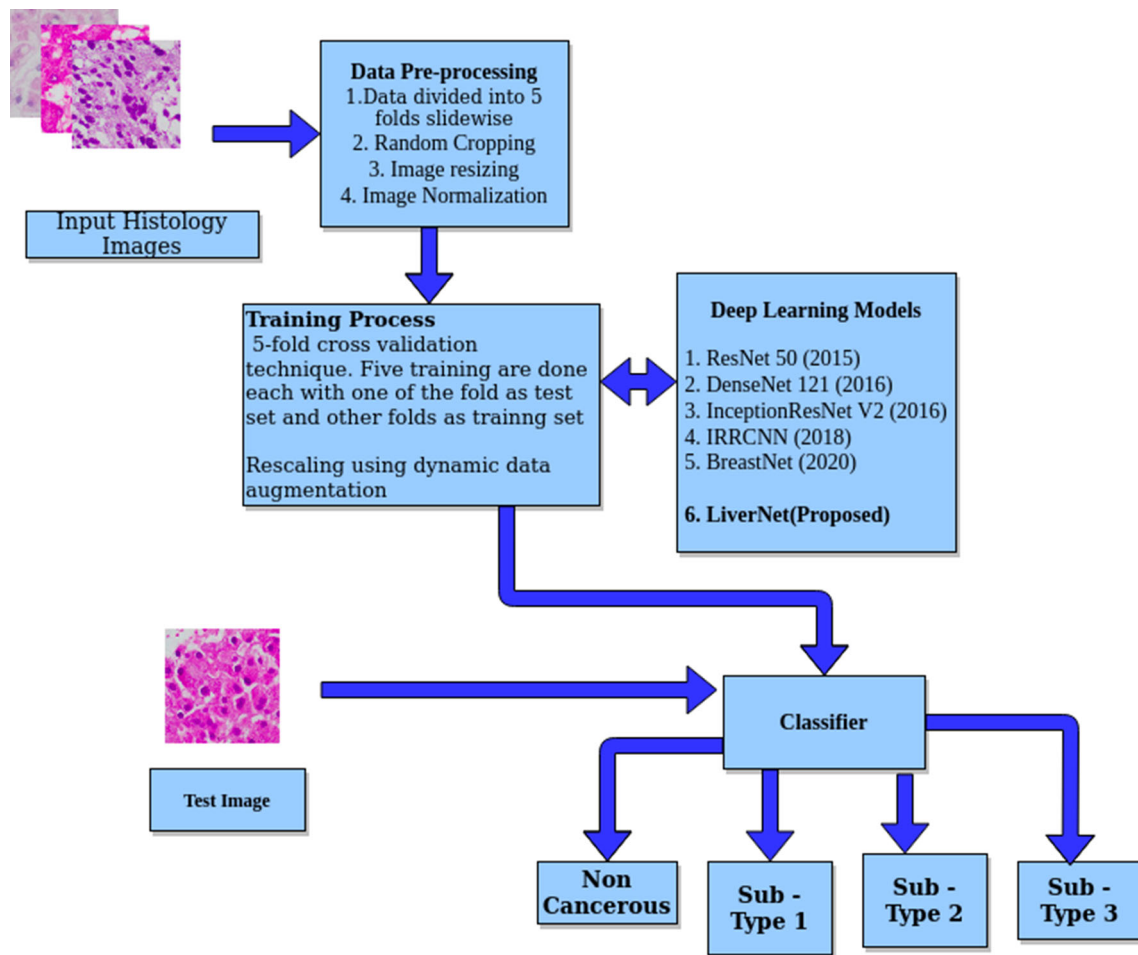


Fig. 2 Proposed methodology

in the input feature map, and the spatial attention block pays attention to **where** the network must focus on.

Consider a feature map $A \in \mathbb{R}^{H \times W \times C}$ input to the CBAM block at an intermediate point in the encoder pipeline, then we can express the action of the CBAM block mathematically as follows:

$$A_c = f_c(A) \cdot A \quad (2)$$

$$A_s = f_s(A_c) \cdot A_c \quad (3)$$

where “.” denotes element-wise multiplication. $f_c : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^{1 \times 1 \times C}$ and $f_s : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^{H \times W \times 1}$ denote the functions of channel-wise and spatial attention blocks, respectively. A_c is the intermediate output following the element-wise multiplication between the channel-wise attention map f_c and the input feature map A . A_s is the final output of the CBAM attention block, and also the product of element-wise multiplication between the spatial attention map and A_c . The channel-wise attention block consists of parallel average- and max-pooling operations that share a common fully connected network before being added. This

can be described given an input A as:

$$f_c = \sigma(FC(AvgPool(A))) + FC(MaxPool(A)) \quad (4)$$

where σ denotes the sigmoid function and FC denotes the shared fully connected layers. The spatial attention block consists of the concatenation of the results of max-pool and average-pool operations before feeding to a convolution layer. This action given an input A can be described as:

$$f_s = \sigma(w \otimes ([AvgPool(A); MaxPool(A)])) \quad (5)$$

where $\otimes$ represents the two-dimensional convolution operation with a kernel w . The residual block used in our proposed architecture similar to the residual block proposed in [9] has been used in [24]. The only difference being the filters of the first convolution layer in our residual block is reduced by a factor of 4 compared to the filter used in the residual block proposed in [24]. This not only reduced the number of parameters required in the model but also improved the features extracted from the input.

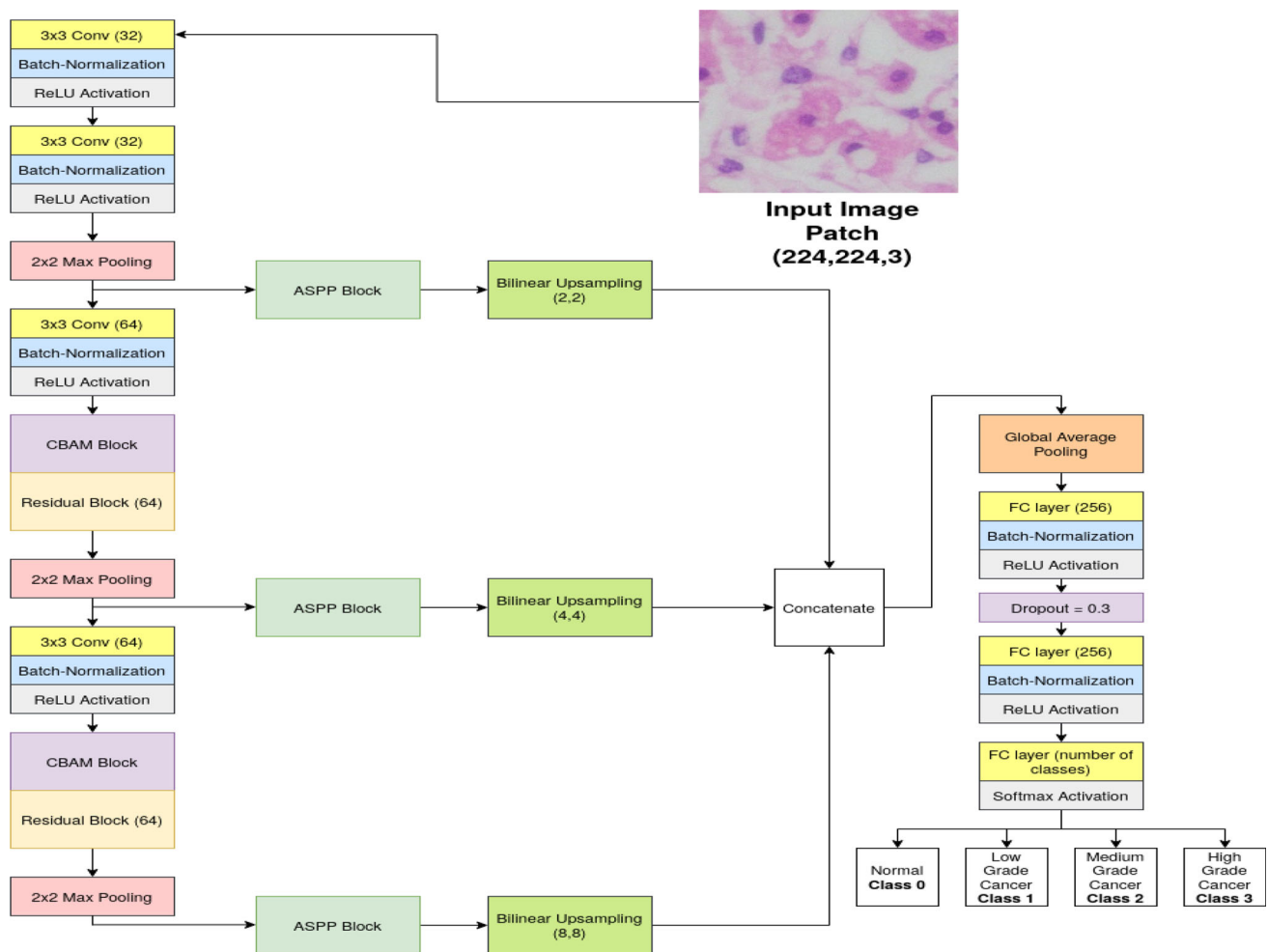


Fig. 3 Proposed architecture

ASPP block

As demonstrated in [25], an atrous spatial pyramid pooling block can successfully extract multi-scale features from a feature map. Considering its effectiveness, we adopt a similar ASPP block in our architecture. Atrous convolution or dilated convolution can be used to expand the receptive field size without increasing the number of parameters involved. Consider an input two-dimensional signal X convolved with a two-dimensional filter w through atrous convolution. Their convolved product can be expressed as:

$$Y[i, j] = \sum_m \sum_n X[i + r.m, j + r.n] w[m][n] \quad (6)$$

Here, r corresponds to the dilation rate or the rate at which the input signal X is sampled. The effect of atrous convolution increasing the receptive field size of the kernel is equivalent to adding $r - 1$ zeros in between the kernel elements. Therefore, in the case of $r = 2$, a 3×3 kernel will

have a receptive field size equivalent to a 5×5 kernel but with only 9 parameters. This is depicted in Fig. 4 for dilation rates $r = 2$ and $r = 3$, and the same concept can be extended for higher dilation rates of $r = 4, 5, 6$, etc.

The ASPP block used in our architecture is shown in Fig. 5. It takes as input a feature map and performs the following operations in parallel before concatenation: (1) 1×1 Convolution. (2) 3×3 Convolution with dilation rate = 2. (3) 3×3 Convolution with dilation rate = 3. (4) 3×3 Convolution with dilation rate = 6. (5) 3×3 Convolution with dilation rate = 8. (6) Global average pooling. Every convolution operation in each branch in this block is followed by a batch-normalization layer and a ReLU activation function.

In our implementation of this block, we have used depthwise separable convolutions to reduce the number of parameters and make the architecture more efficient [13]. This type of convolution shown in [13] requires only $\frac{1}{8}$ th the number of parameters and only $\frac{1}{9}$ th the number of calculations. The expression for depthwise separable convolutions is described by considering an input image I of size $D_f \times D_f \times M$ and

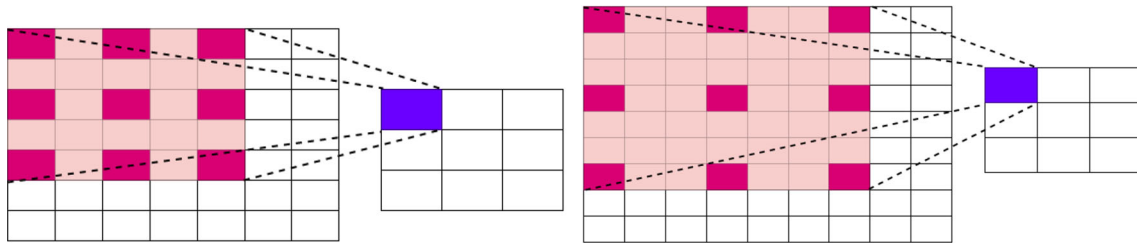


Fig. 4 Atrous convolution for dilation rates 2 (left) and 3 (right)

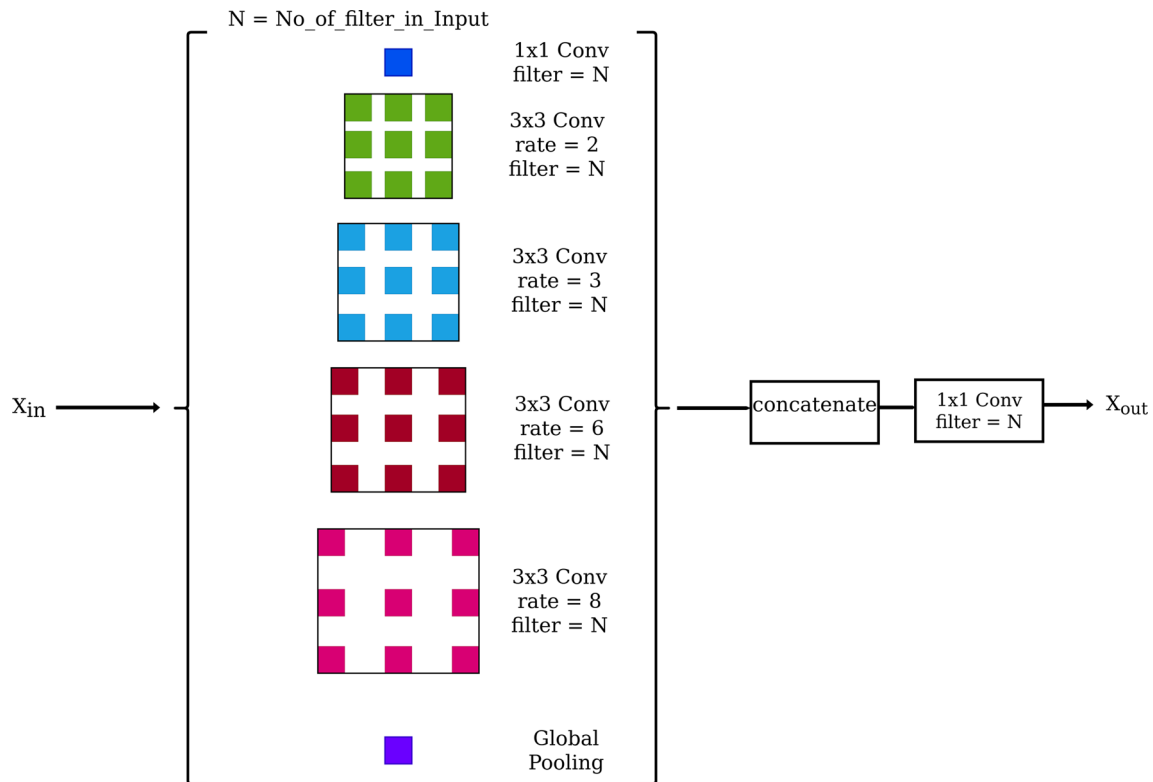


Fig. 5 ASPP block in proposed architecture

a filter w of size $D_k \times D_k \times N$. The depthwise separable convolution can be divided into 2 operations: (1) depthwise convolution and (2) pointwise convolution. In depthwise convolution, I is convolved with a $D_k \times D_k \times 1$ filter along each dimension of I . Hence, at the end of the depthwise convolution process, I would have been convolved with w_D (M filters of size $D_k \times D_k \times 1$).

$$I_D = I \otimes w_D \quad (7)$$

where the size of I_D is $D_p \times D_p \times M$. Following this operation, the pointwise convolution operation uses a $1 \times 1 \times M$ kernel convolved with the intermediate output I_D for each dimension N of the original kernel w . This can be expressed as:

$$I_P = I_D \otimes w_P \quad (8)$$

where w_P is the kernel of size $N \times 1 \times 1 \times M$ and I_P is the final convolution output. All the convolution and pooling outputs are concatenated and passed through a 1×1 convolution layer to maintain the same filter size as that of the input. This convolution output is passed through a batch normalization and ReLU activation layer before they are sent to the bilinear upsampling layer. In our proposed architecture, we place the ASPP block after each max-pooling operation in the encoder pipeline. The motivation for this is that the output of the max-pool layers contains feature-rich information at different extents and scales. ASPP blocks can be used here to extract this information before it is upsampled and processed through the hyper-column technique.

Implementation and training

The proposed architecture was implemented and trained using Python and the keras deep-learning library with tensorflow backend. The Nvidia Tesla K80 GPU speeds up the training process. Prior to the training process, we made sure that the dataset of liver tissue was void of invalid images and labels. Furthermore, we extracted image patches from the original dataset. The dataset was divided into fivefold, and fivefold cross-validation technique was used for the training process. To prevent over-fitting, we re-scaled the dataset using data augmentations techniques, namely flip and rotate. The images were normalized to zero mean and unit variance and resized to 224×224 before feeding into the model. This process has been explained in more detail in the following subsections. L2-regularization was added to the loss function, and the regularization constant λ was set to 0.003. The learning rate α was set to 0.0001, and a learning rate scheduler scheme was employed in the training process which reduced the learning rate by a factor of $e^{-0.1}$ after every epoch to ensure proper convergence. To ensure correctness and an accurate estimation of the performance of the model, we tested all models for five trials on the KMC and TCGA datasets. Each time onefold was used as the test set, and other folds were used as the training set. The results were calculated on fold used as test set, and the final result was obtained by averaging the metrics across the five trials. Each trial consisted of random initialization of the network weights before training it.

Overview of dataset used

To confirm our hypothesis, we have used two datasets, the publicly available TCGA Liver dataset [28] and a corpus of labelled liver tissue images from Kasturba Medical College(KMC), Mangalore, MAHE, Manipal, Karnataka, India. For convenience, we refer to the former dataset as the TCGA-Liver dataset and the latter as the KMC dataset.

TCGA-LIHC

For the preparation of TCGA-Liver dataset [28], a total of 141 slides were obtained from different patients belonging to 3 sub-types of liver HCC tumor—solid tissue normal, primary tumor, and recurrent tumor. The size of each pixel is 0.25×0.25 microns. The normal slides were considered to be sub-type 0, primary tumor sub-type 1, and recurrent tumor sub-type 2. Out of the given slides, 60 slides belong to sub-type 0, 50 slides belong to sub-type 1, and 31 slides are from sub-type 2. These slides were randomly divided into fivefold using K-fold split method and were ensured that no slide was common in any of these folds. Patches of size 1024×1024 were obtained from each of these slides, resulting in around

140 image patch in each of the fold (705 total image patch). Among the 140 patches in each fold, 50 patches belonged to normal type, 60 patches belonged to primary tumor and 30 patches belonged to the secondary tumor.

KMC data

This novel dataset that we have used in this work consists of 257 original slides from different patients each of size 1920×1440 , belonging to 4 sub-types of liver HCC tumor. Each pixel is of size 3.69×3.69 microns. This dataset has been annotated by a clinical pathologist at KMC, Manipal. Out of these samples, 70 slides are from sub-type 0, 80 slides are from sub-type 1, 83 slides are from sub-type 2, and 24 slides are from sub-type 3. Similar to TCGA dataset preparation, these slides were classified into 5 different folds ensuring no slide is common in any of the given folds. Random patches of size 1000×1000 were obtained from each slide. We extracted 5 random patch from each slide in sub-type 0, while 4 patches were extracted from each slide in sub-type 1 and sub-type 2. Fourteen random patches were obtained from each slide in sub-type 3 to remove the imbalance in the dataset. After random patch extraction, we ended up with a total of around 272 image patch per fold (1338 image patch in total). Among the 272 patch, 70 patches belonged to sub-type 0, 64 patches belonged to sub-type 1, 68 patches belonged to sub-type 2, and 70 patches were from sub-type 3.

Overview of training procedure

We made the use of **fivefold cross-validation** technique for training. All the comparison models and the proposed model were trained on the both TCGA dataset and KMC dataset for 5 times each. Each time one of the fivefold was used as the test set, while the rest of the folds were used as the training set. To increase the number of samples as well as to remove any imbalance present in the training data, we conducted dynamic augmentation upon the folds used in the training set. The fold used for the test set was used as it is. For KMC dataset, we used horizontal and vertical flipping for data augmentation, while for TCGA dataset rotation was also used along with flipping operations. For each of the trial, we obtained around 2240 samples for training set and 140 samples in the test set for TCGA case, while 3210 samples in the training set and 272 samples in the test set were obtained and used for the KMC case. The images were resized to a size of 224×224 before feeding into the model. For each turn, the results were calculated on the fold used as the test set. The average value of the metrics across all these training was reported.

Loss function

Let f be the function that best represents the action of the proposed model on an input image. Let $I \in \mathbb{R}^{224 \times 224 \times 3}$ represent an input image and $y \in \mathbb{R}^{K \times 1}$ represent its corresponding class represented in one-hot coded representation. Given an input image I , the classification can be expressed as:

$$\hat{y} = f(I) \quad (9)$$

$\hat{y}$ here is a four element vector representing the softmax probabilities of each class and is an estimation of the classification of I between the K classes. Given a dataset of images and their corresponding labels: $(I_l, y_l)_{l \in [1, N]}$ and weights w , the global loss can be expressed as:

$$L = \frac{1}{N} \sum_{l=1}^N \text{loss}(y_l, \hat{y}_l) + \lambda \|w\|_2^2 \quad (10)$$

where λ is the regularization constant. We have used categorical cross entropy loss between the predicted and ground truth class labels, defined as:

$$\text{loss}(y_l, \hat{y}_l) = - \sum_i^K y_{li} \ln(\hat{y}_{li}) \quad (11)$$

Overview of metrics used for comparison

To compare our proposed model with other competitive image classification models, we use a variety of metrics to qualitatively estimate the performance comparison. The metrics that we have used are precision, recall, F1-score, IoU, and accuracy. In addition to these metrics, we also measure the computational efficiency of our proposed model through the number of parameters required and the number of floating point operations per second.

- (1) **Precision:** This metric measures the proportion of correct positive identifications made against the total positive identifications made.

$$\text{precision} = \frac{TP}{TP + FP} \quad (12)$$

where TP and FP stand for true positives and false positives, respectively.

- (2) **Recall:** This metric measures the proportion of positives that were correctly identified.

$$\text{recall} = \frac{TP}{TP + FN} \quad (13)$$

where FN is false negatives.

- (3) **F1-Score:** The F1-score gives a more balanced view of a model's classification performance. It overcomes the precision–recall trade-off by finding the harmonic mean of the precision and recall metrics.

$$\text{F1-score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (14)$$

- (4) **IoU(JI):** The intersection over union finds the correlation between 2 given sets and helps to understand the similarity between the sets.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (15)$$

In the classification scenario, the IoU has been calculated as the number of correctly classified patches over the total number of selected patches from that class.

- (5) **Accuracy:** Accuracy is the ratio of the number of correct classifications to the total number of predictions made. The formula can be expressed as follows:

$$\text{Accuracy} = \frac{TP + TN}{\text{Total number of pixels}} \quad (16)$$

Results

Performance of the proposed model

In multi-class classification of tissue images into normal, sub-type 1, sub-type 2, and sub-type 3 cancer, our proposed LiverNet model performs excellently well compared to other competitive architectures and outperforms in all metrics under consideration and both datasets used. This is shown in Tables 1 and 2. The training methodology is depicted in Fig. 2, and the box plots of the F1-score for each model are depicted in Fig. 8 for TCGA-Liver and KMC datasets. The class-wise performance of one of the folds is shown by the ROC-AUC plot in Fig. 9. The confusion matrices of fivefold cross-validation for multi-class classification by the proposed model are shown in Figs. 7 and 6 for the KMC and TCGA-Liver datasets, respectively. Through this depiction, the ability of our model to consistently classify HCC histopathology images correctly has been proven. The accuracy of LiverNet was evaluated to 90.93% on the KMC dataset, while the highest accuracy among the models in comparison was 88.41% of the BreastNet architecture [24]. Similarly for the TCGA dataset, the proposed LiverNet model outperforms the BreastNet architecture with an accuracy of 97.72%. In terms of F1-score and IoU, LiverNet showed 90.93% and 83.60%, respectively, on the KMC dataset, and 97.72% and 95.61% on the TCGA-Liver dataset. On the other hand, BreastNet [24] which had the highest

Table 1 Comparison of the proposed LiverNet model with other competitive architectures on the test set of the KMC dataset

Metrics	Class	ResNet (2015)	DenseNet (2016)	InceptionResNetV2 (2016)	IRRCNN (2018)	BreastNet (2020)	LiverNet (proposed)
Precision	Type 0	0.9809	0.9759	0.9727	0.9972	0.9891	0.9944
	Type 1	0.8255	0.8129	0.7676	0.8032	0.8496	0.8601
	Type 2	0.7202	0.7534	0.7853	0.7429	0.7394	0.7945
	Type 3	0.9964	0.9809	0.9929	0.9665	0.9898	0.9972
	Average	0.8652	0.8651	0.8677	0.8748	0.8841	0.9093
Recall	Type 0	1.0000	0.9943	1.0000	1.0000	1.0000	1.0000
	Type 1	0.7094	0.8156	0.7781	0.7531	0.775	0.825
	Type 2	0.8658	0.8096	0.8279	0.8204	0.8616	0.8638
	Type 3	0.8771	0.8371	0.8571	0.9171	0.8914	0.94
	Average	0.8652	0.8651	0.8677	0.8748	0.8841	0.9093
F1-score	Type 0	0.9903	0.9846	0.9857	0.9986	0.9945	0.9972
	Type 1	0.7536	0.8124	0.7632	0.7757	0.8091	0.8405
	Type 2	0.7783	0.768	0.8047	0.7766	0.7942	0.8265
	Type 3	0.9246	0.882	0.9073	0.9337	0.9323	0.9633
	Average	0.8652	0.8651	0.8677	0.8748	0.8841	0.9093
IOU(JI)	Type 0	0.9809	0.9701	0.9727	0.9972	0.9891	0.9944
	Type 1	0.6112	0.688	0.6227	0.6361	0.6884	0.7328
	Type 2	0.6422	0.6339	0.6757	0.6374	0.6696	0.7111
	Type 3	0.8749	0.8231	0.8529	0.8925	0.8849	0.9372
	Average	0.766	0.7674	0.7712	0.7793	0.7973	0.836
Accuracy	Type 0	0.9948	0.9917	0.992	0.9993	0.997	0.9985
	Type 1	0.8925	0.9088	0.8809	0.8963	0.9136	0.9264
	Type 2	0.8756	0.8764	0.901	0.883	0.8878	0.91
	Type 3	0.9675	0.9534	0.9616	0.971	0.9698	0.9836
	Average	0.8652	0.8651	0.8677	0.8748	0.8841	0.9093
Parameters	–	23,591,810	7,039,554	54,339,810	6,077,059	605,823	573,936
FLOPs	–	47,024,420	13,912,175	108,531,281	12,117,656	1,204,845	1,193,484

Table 2 Comparison of the proposed LiverNet model with other competitive architectures on the test set of the TCGA-Liver dataset

Metrics	Class	ResNet (2015)	DenseNet (2016)	InceptionResNetV2 (2016)	IRRCNN (2018)	BreastNet (2020)	LiverNet (proposed)
Precision	Type 0	0.9516	0.951	0.9773	0.9608	0.9884	0.9923
	Type 1	0.9266	0.9686	0.9711	0.9756	0.9695	0.9965
	Type 2	0.8873	0.8902	0.8527	0.9315	0.9081	0.9239
	Average	0.9205	0.9308	0.9407	0.9588	0.9604	0.9772
Recall	Type 0	0.996	0.984	0.992	0.992	1.0000	1.0000
	Type 1	0.89	0.87	0.8867	0.9267	0.9367	0.95
	Type 2	0.8552	0.961	0.961	0.9686	0.9438	0.9933
	Average	0.9205	0.9308	0.9407	0.9588	0.9604	0.9772
F1-score	Type 0	0.973	0.9647	0.9843	0.9752	0.9941	0.9961
	Type 1	0.9063	0.9091	0.926	0.9498	0.9521	0.9723
	Type 2	0.8548	0.9177	0.9015	0.9495	0.9234	0.9567
	Average	0.9205	0.9308	0.9407	0.9588	0.9604	0.9772
IOU(JI)	Type 0	0.9479	0.935	0.9693	0.9528	0.9884	0.9923
	Type 1	0.8309	0.8409	0.8634	0.9054	0.9099	0.9469
	Type 2	0.7554	0.8542	0.8225	0.9041	0.8613	0.9187
	Average	0.8548	0.8746	0.8886	0.9215	0.9247	0.9561
Accuracy	Type 0	0.98	0.9729	0.9886	0.9815	0.9958	0.9971
	Type 1	0.9205	0.9308	0.9407	0.9588	0.9604	0.9772
	Type 2	0.9405	0.958	0.9521	0.9773	0.9646	0.9801
	Average	0.9205	0.9308	0.9407	0.9588	0.9604	0.9772
Parameters	–	23,591,810	7,039,554	54,339,810	6,077,059	605,823	573,936
FLOPs	–	47,024,420	13,912,175	108,531,281	12,117,656	1,204,845	1,193,484

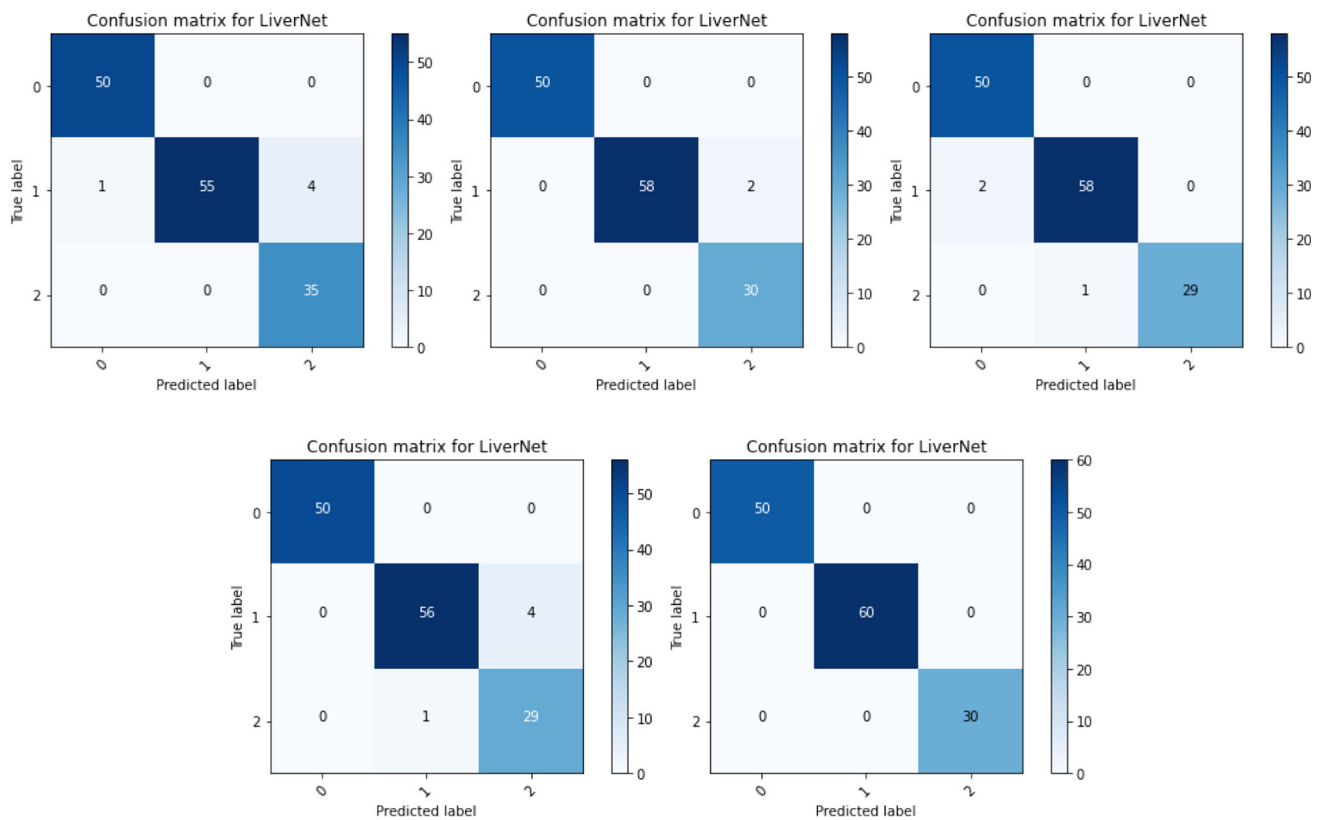


Fig. 6 Confusion matrix for LiverNet generated for the fivefold cross-validation on the TCGA dataset (fold 1 to 5 used as test set from left to right)

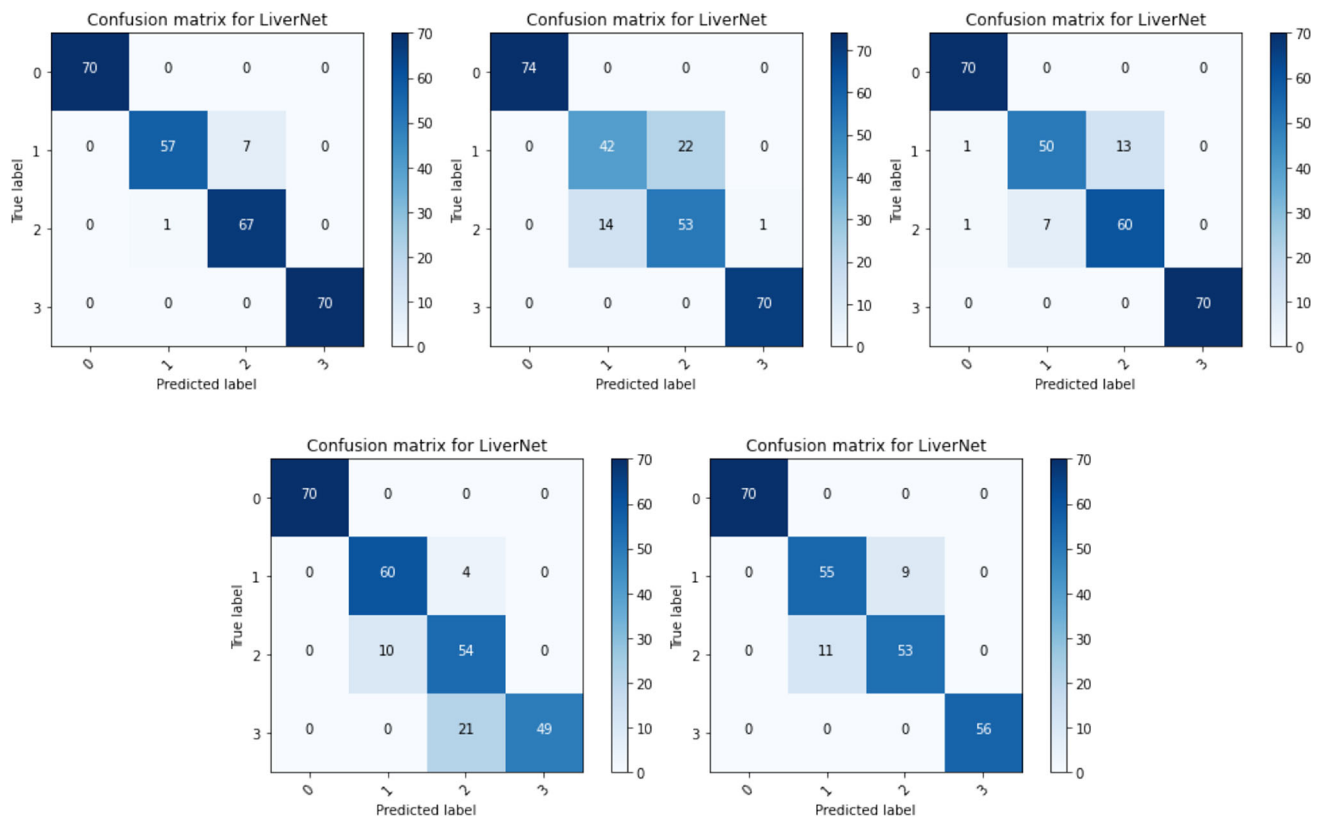


Fig. 7 Confusion matrix for LiverNet generated for the fivefold cross-validation on the KMC dataset (fold 1 to 5 used as test set from left to right)

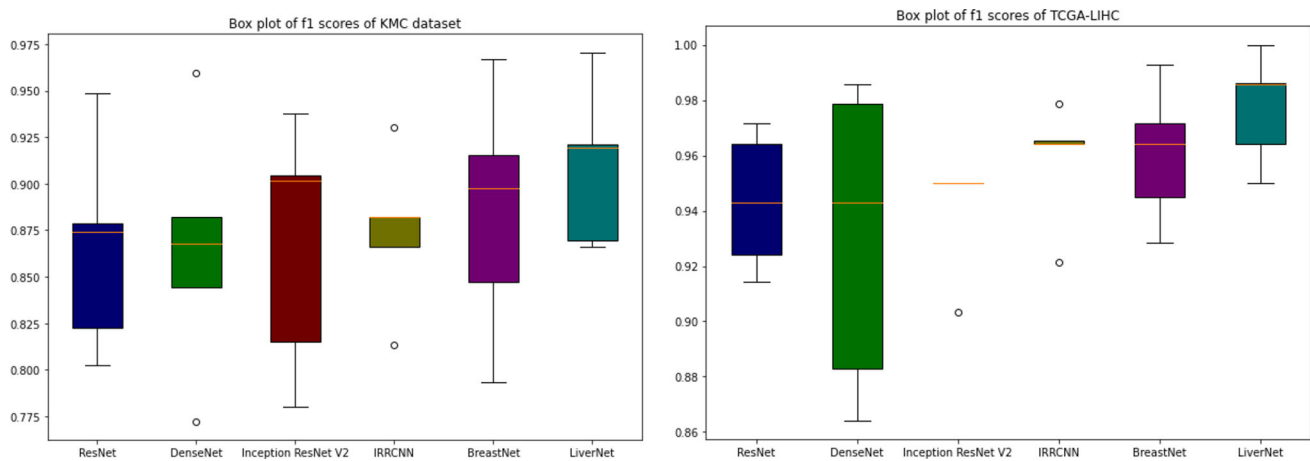


Fig. 8 Box plots for the F1-score on the KMC (left) and TCGA-LIHC (right) datasets

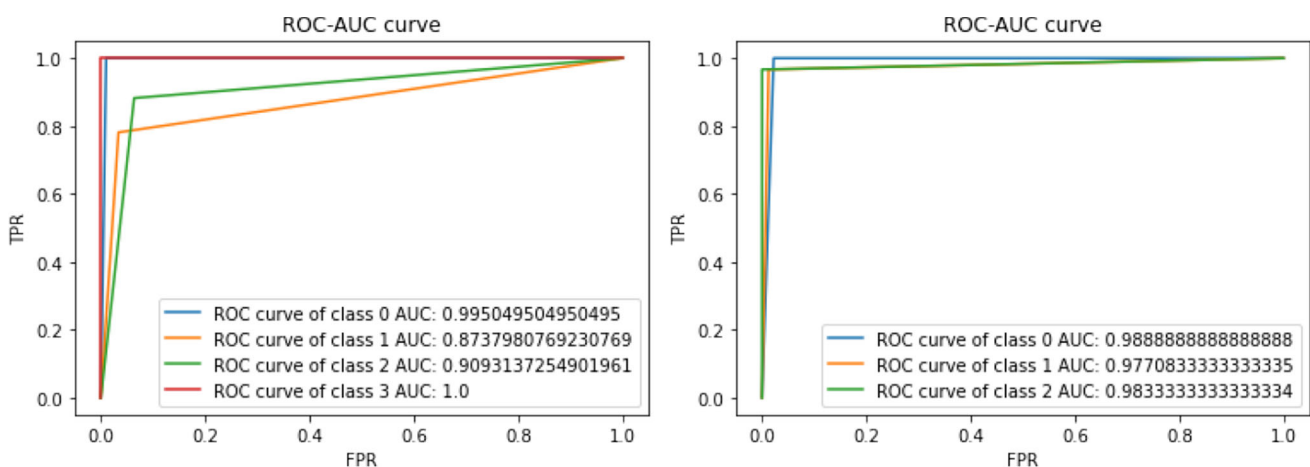


Fig. 9 ROC-AUC curves for fold 3 of KMC (left) and TCGA-LIHC (right) datasets

score among compared models got an average F1 of 88.41% and IoU of 79.73%, respectively, on the KMC dataset and an average F1 of 96.04% and IoU of 92.42%, respectively, on the TCGA-Liver dataset. “Micro” average was used for the calculation of precision, recall, F1-score, and IoU for this multi-class classification experiment.

Our LiverNet model shows a profound improvement in classification of liver histopathology images into respective sub-types on the KMC and TCGA-Liver datasets. We attribute this improvement to the ASPP block that has the ability to extract information at multiple scales from a feature map, for further processing. The placement of ASPP blocks after each intermediate max-pool operation can help derive more relevant features for classification. In terms of computational efficiency, we attribute this to the breastnet architecture [24], which prefers to use simple bilinear upsampling, compared against transpose convolutions which will additionally require millions of parameters if used.

Computational complexity

It can be observed in Tables 1 and 2 that our proposed model outperforms all the other models in the criteria of computational efficiency. The comparison can be visualized better in Fig. 10. The proposed LiverNet model exhibits the lowest number of parameters and floating point operations per second (FLOPs) compared to all other models including BreastNet. This can be attributed to the reduction of number of filter by a factor of 4 in the first convolution layer in each residual block. This not only helped in reducing the parameters but also improved the performance of architecture. Depthwise-separable convolutions used in all ASPP blocks are another reason for efficiency of the model. Depthwise separable convolutions help reduce the number of parameters by performing channel-wise and spatial convolutions as two separate operations, thereby reducing the total number of parameters in the kernel. This can also be used to explain the reduction in number of FLOPs. In addition to this, we have incorporated bilinear-upsampling operations and have used

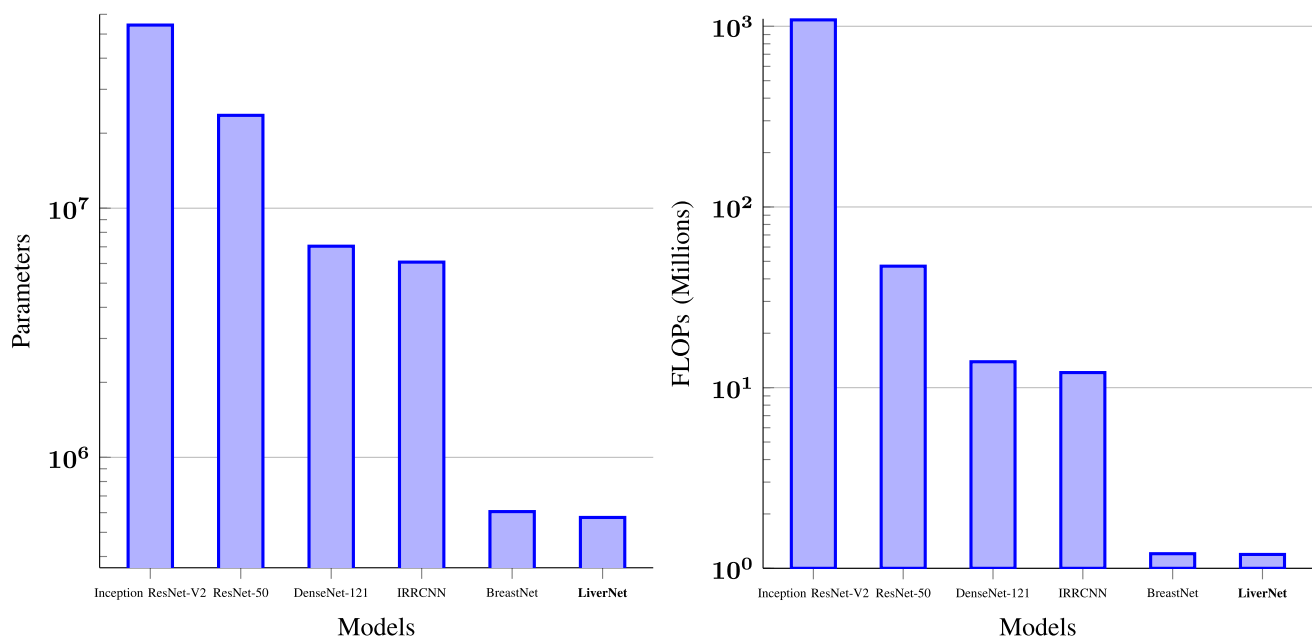


Fig. 10 Computational complexity of different architectures: (left) comparison of number of parameter, (right) comparison of number of flops

this in preference to transpose convolutions, as the latter will require a high number of parameters and FLOPs. Further, an increase in these metrics is observed over [24] due to the additional ASPP blocks added.

Conclusion

Through this research study, we have introduced a novel and extremely effective deep learning architecture to classify liver tissue images between multi-class labels of normal, low sub-type liver HCC tumor, medium sub-type liver HCC tumor, and high sub-type liver HCC tumor. Having conducted extensive experiments on liver histopathology image data, we introduce the LiverNet framework, which can serve as a benchmark for future endeavors in this field. Our empirical results demonstrated that our architecture outperforms previous state-of-the-art models in liver histopathology image sub-type classification by a respectable margin with an accuracy of 90.93%, precision of 0.9093, recall of 0.9093, F1-score of 0.9093, and IoU of 0.8360 on the KMC dataset. Similarly, improvements can be seen on the TCGA-Liver dataset with an accuracy of 97.72%, precision of 0.9772, recall of 0.9772, F1-score of 0.9772, and IoU of 0.9561. In addition to the improvement in quality of classification, our model is also incredibly efficient requiring a fraction of the number of parameters and FLOPs that previous state-of-the-art architectures require. We hope that this research study will spread awareness and interest into the domain of liver cancer diagnosis and treatment planning. A limitation of this study is that the dataset we have used is restricted in terms

of size and variance. This limitation is majorly attributed to the deficiency in liver histopathology image data available in the public sphere. This limitation can be overcome by using a dataset that encompasses a large number and diverse set of patients with HCC-related problems.

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Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval KMC Liver histopathology dataset was used for experimentation. Ethical approval has been obtained from Kasturba Medical College (KMC) Mangalore, Manipal Academy of Higher Education (MAHE), Manipal, Karnataka, India. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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